

The Big Wheel at $z = 3.25$: A Clean Galaxy-Scale Rotation Curve Disfavors a Rising MOND Acceleration Scale

Carl P. Zimmerman (Briar Creek Tech) 2026-07-05. One backing script, exit 0, both coefficient footings: `real_research/reviews/bigwheel_a0z_2026/bigwheel_a0z_definitive.py`.

Abstract

The MOND acceleration scale a_0 has been proposed to track the dark-energy density, $a_0 \propto \sqrt{\rho_{\text{DE}}}$ (Limbach, Psaltis & Özel 2008; Milgrom 1999), which makes it *evolve* with redshift and turns high- z galaxy kinematics into a cosmological probe. Competing proposals instead tie a_0 to the Hubble rate, $a_0 \propto cH(z)$ (entropic/emergent-gravity and quantized-inertia constructions), which makes a_0 *rise* at high z . The two readings diverge by a factor ≈ 5 by $z \approx 3$. We confront these with the first clean galaxy-scale probe at that epoch: **the Big Wheel** ($z = 3.25$, Nature Astronomy 2025, arXiv:2409.17956), an exceptionally large, rotation-dominated ($V/\sigma \approx 4.5$) disk whose deep-MOND baryonic Tully–Fisher relation reads the acceleration scale straight off the rotation curve, $a_{0,\text{eff}} = V_{\text{circ}}^4/(G M_{\text{bar}})$. With a Monte-Carlo propagation of the velocity and mass errors, a dynamical-mass bound on the (over-estimated) SED stellar mass, and a gas-only hard ceiling, we find **$a_{0,\text{eff}} \approx 1.0\text{--}1.6 \times 10^{-10} \text{ m s}^{-2}$ — its local value**. The rising branch, which requires $a_0(3.25) \approx 4.6\text{--}5.9 \times 10^{-10}$ (a disk spinning at 400–425 km s $^{-1}$), is **disfavored at $\approx 2\sigma$** — the probability that the measured $a_{0,\text{eff}}$ reaches the rising value is only $P \approx 1\text{--}3\%$ (the disk spins at 280 ± 31 km s $^{-1}$, not the required 400–425); a **constant** a_0 (268–285 km s $^{-1}$ predicted) is an excellent match; and the **declining** $\sqrt{\rho_{\text{DE}}}$ branch (246–262 km s $^{-1}$) is consistent, sitting $\approx 1\sigma$ low. Both coefficient footings (framework canonical $a_0 = 9.36 \times 10^{-11}$; empirical SPARC $a_0 = 1.20 \times 10^{-10}$) give the same verdict. The result is reported straight, with its single-object systematics foregrounded and its tension with the MUSE-DARK III statistical “rising” signal stated. The clean conclusion is a *negative* one that nonetheless matters: **the best high- z disk available disfavors a strongly rising acceleration scale and is consistent with a near-constant one** — favoring the $\sqrt{\rho_{\text{DE}}}$ reading over the cH reading, while not yet detecting the distinctive high- z decline that $\sqrt{\rho_{\text{DE}}} +$ evolving dark energy predicts.

1. The test

In the deep-MOND regime the baryonic Tully–Fisher relation is exact, $V_{\text{flat}}^4 = G M_{\text{bar}} a_0$, so a single well-measured rotation curve reads the acceleration scale directly:

$$a_{0,\text{eff}} = \frac{V_{\text{circ}}^4}{G M_{\text{bar}}}.$$

At $z = 3.25$ the candidate laws for a_0 diverge sharply. Writing $a_{0,\text{loc}}$ for the local value:

- **H_RISE** — $a_0(z) = a_{0,\text{loc}} \cdot E(z)$, with $E(z) = \sqrt{(\Omega_m(1+z)^3 + \Omega_\Lambda)}$. This is the $a_0 \propto cH(z)$ family (entropic gravity à la Verlinde; McCulloch’s quantized inertia, $a_0 = cH_0/6$; the Ho-Minic-Ng Modified-Dark-Matter scale). $E(3.25) = 4.95$, so a_0 rises $\approx 5\times$.

- **H_CONST** — $a_0(z) = a_{0_loc}$. Ordinary MOND, and the framework’s safe core if the dark energy is a true cosmological constant ($w = -1$, ρ_{DE} constant $\Rightarrow a_0$ constant).
- **H_DECL** — $a_0(z) = a_{0_loc} \cdot \sqrt{\rho_{DE}(z)/\rho_{DE0}}$. The framework’s distinctive reading: $a_0 \propto \sqrt{\rho_{DE}}$ (LPO 2008) evaluated on DESI DR2 evolving dark energy (CPL $w_0 = -0.75$, $w_a = -0.86$), for which $\rho_{DE}(3.25)/\rho_{DE0} = 0.51 \Rightarrow a_0$ declines to $0.71 \times$ local.

A single rotation curve at $z = 3.25$ therefore separates a factor $\approx 5 / 0.71 \approx 7$ in a_0 between the rising and declining extremes — a wide, clean lever.

2. The datum, and its systematics

The Big Wheel (arXiv:2409.17956) is the cleanest galaxy-scale target available at this epoch: an unusually large disk, rotation-dominated ($V/\sigma \approx 4.5$), with a resolved rotation curve and a CO-traced molecular gas mass.

- **V_rot** = **280 (+32/−31) km s^{−1}** (inclination-corrected maximum at $R_{eff} = 9.6$ kpc, inclination $\approx 50^\circ$). We apply a small asymmetric-drift correction $V_{circ}^2 = V_{rot}^2 + 3.4 \sigma^2$ ($\sigma_{int} = 61$ km s^{−1}), which the high V/σ keeps modest.
- **M_gas(H₂)** = **1.8 (+1.0/−0.8) × 10¹¹ M_⊙** (CO), with a 1.36 helium correction.
- **M_star**: the SED value $3.7 \times 10^{11} M_{\odot}$ is over-estimated — the dynamical mass within R_{eff} , $M_{dyn} = V^2 R / G = 1.75 \times 10^{11} M_{\odot}$, is *smaller than the SED stellar mass alone*, which is unphysical ($M_{bar} \leq M_{dyn}$). We therefore adopt a parametric $M_{star} = 1.7 \times 10^{11} M_{\odot}$ as the fair central case, and separately quote a hard **gas-only** upper bound on a_{0_eff} ($M_{bar} = 1.36 M_{gas}$, no stars).

These are the honest single-object systematics; each is bounded rather than assumed away.

3. Results (both footings)

Monte-Carlo (4×10^5 draws) over the velocity and mass errors gives the measured scale:

M_bar assumption	$a_{0_eff} (10^{-10} \text{ m s}^{-2})$
SED ($M_{star} = 3.7 \times 10^{11}$, over-estimated)	1.03 (+0.60/−0.38)
Parametric ($M_{star} = 1.7 \times 10^{11}$, fair central)	1.54 (+1.07/−0.60)
Gas-only (hard upper bound)	≤ 2.55

So **$a_{0_eff} \approx 1.0\text{--}1.6 \times 10^{-10} \text{ m s}^{-2}$** — **consistent with the local value on either footing** (framework 0.94; empirical 1.20). The three hypotheses predict, for the fair-central baryonic mass:

		$a_0(3.25) [10^{-10}]$	$V_{\text{pred}} [\text{km s}^{-1}]$	vs measured 280 ± 31
H_RISE	(frame-work / empirical)	4.63 / 5.94	400 / 425	disfavored $\approx 2\sigma$ ($P \approx 1\text{--}3\%$)
H_CONST		0.94 / 1.20	268 / 285	match
H_DECL		0.67 / 0.86	246 / 262	$\approx 1\sigma$ low, consistent (undetected)

The rising branch is disfavored at $\approx 1.9\sigma$ (framework footing) to $\approx 2.3\sigma$ (empirical footing): the probability that the measured a_0 eff reaches the rising value is $P \approx 1\text{--}3\%$. (A naive median/ σ ratio reads higher, $\approx 2.9\text{--}4.1\sigma$, but a_0 eff = $V^4/(G M_{\text{bar}})$ has a heavy upper tail, so that ratio over-counts; the tail probability, which correctly folds the full M_{bar} systematic into the prediction as well as the measurement, is the honest number — and it is invariant, giving the same $\approx 2\sigma$ whether computed in a_0 or in velocity space.) In velocity space the statement is blunt: a fivefold-risen a_0 would spin this baryon load at $400\text{--}425 \text{ km s}^{-1}$, and it spins at 280 — but the $\approx 2\times$ baryonic-mass systematic widens the rising *prediction* to $\pm \sim 20\%$ as well, which is why the honest significance is $\approx 2\sigma$, not the naive 4σ . A constant a_0 predicts $268\text{--}285 \text{ km s}^{-1}$ and lands on the measurement. The declining $\sqrt{\rho}_{\text{DE}}$ branch predicts $246\text{--}262 \text{ km s}^{-1}$ — low by $\approx 6\text{--}12\%$, i.e. within $\approx 1\sigma$ of the $\pm 11\%$ velocity error, and thus consistent but undetected.

4. What this does and does not show

Does (clean): the best clean galaxy-scale disk at $z \approx 3$ *disfavors a strongly rising acceleration scale* at $\approx 2\sigma$ ($P \approx 1\text{--}3\%$). This is a real, if modest and negative, constraint on the $a_0 \propto cH(z)$ family — entropic gravity, quantized inertia, and the rising reading of the framework’s own coefficient fork. It also favors the framework’s $\sqrt{\rho}_{\text{DE}}$ footing (canonical 9.36×10^{-11}) over the cH footing, an internal discriminant. The constraint is one galaxy at $\approx 2\sigma$, not a decisive exclusion; a sample is needed to reach $3\sigma+$.

Does not (honest): it does **not** confirm the framework’s *distinctive* prediction. The distinctive, novel content — a_0 declining $\propto \sqrt{\rho}_{\text{DE}}$ at high z — predicts $246\text{--}262 \text{ km s}^{-1}$, only $\approx 1\sigma$ below the measurement. One galaxy with an $\pm 11\%$ velocity error cannot yet separate a mild decline from a constant. The datum is therefore equally consistent with **ordinary constant- a_0 MOND**; it does not require the evolution. If anything the central value sits slightly *above* the declining prediction, so the data mildly prefer constant over strong decline — comfortable for the $w = -1$ core, undemonstrative for the evolving-DE signature.

In tension with: the MUSE-DARK III statistical stacking analysis (Ciocan et al. 2026), which reports a_0 *rising* with redshift. The Big Wheel — the most direct, gas-traced, single-object probe — points the opposite way. The two need not be reconciled here; a stacked TFR intercept and a resolved rotation curve carry different systematics (baryon-fraction and M/L degeneracies for the former; inclination and single-object mass for the latter). The honest reading is that the high- z a_0 evolution is observationally *contested and not yet settled*, and the cleanest single disk disfavors the rise.

5. Caveats, stated plainly

- **One galaxy.** This is a single rotation curve; the $a_0(z)$ question is decided by a *sample* of clean, gas-traced $z \approx 3$ disks, of which the Big Wheel is the first. No statistical claim about the population is made.
- **Baryonic-mass systematics** dominate the error and are bounded, not eliminated: the SED stellar mass is demonstrably over-estimated (M_{dyn}), the parametric value is a fair central choice, and the gas-only ceiling is a hard bound. The rising branch is disfavored for the SED and parametric (physical) masses; in the ultra-conservative gas-only case (baryons at their hard minimum, $a_{0,\text{eff}}$ at its ceiling $\approx 2.6 \times 10^{-10}$) the rise is only $\approx 1\sigma$ away ($P \approx 15\text{--}22\%$) and is *not* excluded — the $\approx 2\sigma$ disfavoring rests on the physical masses. Note the M_{bar} systematic ($\approx 2\times$) is *itself larger than* the $0.71\times$ decline signal, so this datum cannot probe the declining branch even in principle; it has leverage on the $5\times$ rise only because that gap survives the systematic.
- **Pressure support** is corrected (asymmetric drift, $V/\sigma = 4.5$, small); inclination ($\approx 50^\circ$) enters the velocity.
- **Not a detection of evolution.** The result is a rejection of the rise and a consistency with constant/mild-decline, not a measurement of $a_0(z)$.

6. Context and priority

The relation $a_0 \propto \sqrt{\rho_{\text{DE}}}$ and its redshift test are due to **Limbach, Psaltis & Özel (2008)** (their Eq. 4); the deep-MOND vacuum-scale reading traces to **Milgrom (1999)**. The rising alternative $a_0 \propto cH(z)$ appears in **Verlinde (2010)** entropic gravity, **McCulloch's** quantized inertia ($a_0 = cH_0/6$), and the **Ho-Minic-Ng** Modified-Dark-Matter program — all of which the Big Wheel constrains. This note contributes only the *confrontation of a new clean datum* with these standing hypotheses; the framework's non-monotonic $a_0(z)$ reading and its coefficient are set out separately (Zenodo 10.5281/zenodo.21110936; 10.5281/zenodo.20938891). Nothing here derives a_0 ; it is a data note, reported straight, and its headline is a null that happens to cut cleanly against the rivals.

Script

real_research/reviews/bigwheel_a0z_2026/bigwheel_a0z_definitive.py (exit 0): Monte-Carlo $a_{0,\text{eff}}$ on both mass assumptions and the gas-only bound; the three hypotheses on both coefficient footings, in a_0 and in velocity space; and the significance of the rising-branch rejection. All numbers in this note are its output.

References

- Limbach, S. A., Psaltis, D. & Özel, F. (2008), arXiv:0809.2790.
- Milgrom, M. (1999), Phys. Lett. A 253, 273.
- Big Wheel discovery paper, arXiv:2409.17956 (Nature Astronomy 2025).
- Ciocan, B. et al. (MUSE-DARK III, 2026).
- Verlinde, E. (2010), arXiv:1001.0785; McCulloch, M. (2007), arXiv:0806.4159; Ho, Minic & Ng, arXiv:1601.00662.